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Multi-objective optimization of multi-energy complementary integrated energy systems considering load prediction and renewable energy production uncertainties

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A B S T R A C T

Multi-energy complementary integrated energy system (MCIES) is considered as a promising solution to mitigate carbon emissions and promote carbon peaking and carbon neutrality. Currently, the capacities of a MCIES are sized according to the deterministic load and parameters of the system model. However, uncertainty may lead to the failure to achieve the desired performance and affect the sizing of the MCIES. This study explored an optimization model for the proper sizing of the MCIES considering uncertainties to achieve the best economic, environmental and thermal comfort benefits. The non-dominated sorting genetic algorithm-II (NSGA-II) combined with a technique for order preference by similarity to an ideal solution (TOPSIS) and Shannon entropy method were adopted to solve the optimization. Case studies, an actual swimming pool building with MCIES, as the prototype, were used to illustrate the procedure. Moreover, the effects of uncertainty degree and scenario setting were investigated. The results show the benefits of the proposed approach against the traditional deterministic optimization method for comprehensive consideration of economy, environment and thermal comfort. It also suggests that uncertainty and scenario setting should be carefully and properly considered during the design stage, as they have a significant impact on the results of sizing.

1. Introduction

Energy consumption and carbon emissions in the building sector have accounted for 46.5% of total energy consumption and 51.2% of energy carbon emissions in China, respectively [1]. The utilization and exploitation of renewable energy is the key to save energy and reduce carbon emissions in the buildings sector [2]. However, renewable energy sources are of intermittent and unstable nature. Combining the greenness of renewable energy sources with the stability of conventional energy sources is an effective solution to the problem.

Multi-energy complementary integrated energy system (MCIES) has garnered significant attention as it represents a valuable way for exploiting renewable energy sources with conventional energy sources. The MCIES can utilize multiple energy sources [3], which can flexibly be transformed into the energy type that users can directly consume through technological conversion [4,5]. Recent studies show the significant benefits of MCIES. For example, Ma et al. [6] conducted a study of a multi-energy complementary heating system, and the results showed that the energy-saving rate is about 30% compared with the conventional central heating system. Ye et al. [7] found that the multi-energy complementary system can promote the rational development of the type of energy use and the institutional improvement of the clean environmental energy industry.

To realize the expected potential of MCIES, however, a proper design is very important, especially in rationalizing the system configurations and sizes. Note that the MCIES optimal design is a complex task involving multiple objectives [8], such as an economic (annual cost), thermal comfort (load satisfaction rate) and environmental (emission reduction) objective function. Currently, multi-objective optimization methods generally focus on a

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weighted combination of several objectives into a single objective function [9]. While the weighted sum method simplifies the calculation, it does not facilitate the judgment of the decision-makers because it is difficult to correctly correlate several objectives with different natures [10]. For example, the goal of achieving the lowest cost may conflict with environmental protection as well as user-side thermal comfort requirements. Moreover, unlike a single-objective function problem with a unique optimal solution, the solution of the multi-objective function is a solution set containing multiple non-differentiable equilibrium solutions, which is called the optimal non-inferior solution set or Pareto solution set [11]. Pareto solution set can be generated by the non-dominated sorting genetic algorithm-II (NSGA-II) [12]. NSGA-II is one of the most popular optimization algorithms with high running speed, good solving convergence and low computational complexity, which is widely applied in energy system optimization researches [13]. Mei et al. [14] effectively solved the Pareto front of the multi-objective optimization problem for integrated energy systems by applying the improved NSGA-II method. Zhang et al. [15] constructed a comprehensive evaluation index for distributed multi-energy complementary energy systems with the objective of optimal economy, environmental protection and energy efficiency by NSGA-II.

While the theoretical performance of MCIES is impressive, the uncertainty factor could hinder the deterministic multi-objective optimal design methods' practical application. Uncertainty is defined as the information gap between the current information state of the decision-maker and the reality [16,17]. For example, in the traditional design process, load prediction is usually performed using meteorological parameters of the typical year. However, the actual meteorological parameters in the future, such as outdoor ambient temperature and solar radiation, are impossible the same as that used in the design stage. Uncertainty is inevitable during the system design process. The uncertainty sources for optimal design originate from load prediction, e.g. cooling/heating and electrical load, and renewable energy production [18]. When performing load calculations, there are many uncertain parameters, such as occupant [19] and environmental parameters, which lead to uncertainty in load prediction. Uncertainties in renewable energy production are subject to many exogenous conditions, e.g. solar irradiation, wind speed, and outdoor temperature, etc. The profile of these uncertainties, if not properly considered when determining the design, can lead to poor performance [20], all in terms of economic, comfort and environmental objectives. However, the existing deterministic optimal design methods usually yield a smaller margin of system performance [21]. In contrast, considering uncertainties of energy demand can effectively avoid improper sizing [22,23]. Some methods have been developed to address the uncertainties for the optimization problem, e.g. robust optimization [24], and stochastic optimization [25]. The robust optimization method gives the set of uncertain parameters and seeks a solution that performs well for all implementations, but suffers from the problem of over-conservative decision making [26]. Stochastic optimization expresses the uncertainty of parameters as a probability density function to get the best or worst performance expected from the system, widely used for uncertainty-optimized design [27]. Wang et al. [28] obtained the uncertainty of peak load prediction for HVAC design by simulating the probability distribution of uncertain parameters with a large number of samples. Xiao et al. [29] considered the stochasticity of distributed power output and load based on the full probability formula. Although the full probability formula is effective to address the uncertainty problem, the drawback is the unaffordable computational burden, especially for complex systems with various uncertain parameters.

The scenario-based uncertainty analysis method can effectively reduce the calculation compared to the full probability models [30]. Wang et al. [31] set up 14 load demand scenarios based on heating and cooling loads when optimizing the uncertainty distributed energy system design. Zhu et al. [32] selected 24 scenarios considering four renewable energy scarcity levels and two restriction violation levels to minimize total system cost. Yin et al. [33] divided the wind power operation status of five scenarios according to the wind speed to reduce the impact of the volatility of the wind turbine output on the distribution network. However, the above studies perform scenario partitioning considering renewable energy production or load uncertainties separately, less considering both. And the impact of the scenario partitioning approach on optimization results is rarely discussed.

Based on the above, this paper proposed a multi-objective stochastic optimization model for the proper sizing of the MCIES considering load prediction and renewable energy production uncertainties. Moreover, we designed five uncertainty degrees and compared the optimization results under different uncertainty degrees. Additionally, the effects of scenario setting were investigated. The proposed uncertainty-based multi-objective optimization method could be carried out in two steps. First, the uncertain parameters were selected and quantified by the probability distribution. The load prediction can be acquired by the Monte Carlo simulation. Second, the uncertain input data were introduced to NSGA-II to obtain the Pareto solution set. TOPSIS combined with Shannon entropy was employed to select the optimal solution from the Pareto solution set.

The innovations of this paper include the following:

- Combining uncertainty analysis with NSGA-II to obtain the maximum economic, environmental and thermal comfort benefits of the MCIES.
- The impacts of uncertainty degree on the optimal solution were quantified.
- The effects of scenario setting on optimization objectives and operating results of MCIES were investigated.

The remainder of this paper is organized as follows. Section 2 presents the stochastic optimization methods and the optimization objectives. Section 3 demonstrates the optimization method through a case study and provides the discussions and analysis of the results. And Section 4 draws the main conclusions and future works.

2. Methods

Fig. 1 shows the framework of the proposed uncertainty-based multi-objective capacity optimization method for the MCIES, which can be divided into four parts: (1) Model of the system; (2) Uncertainty quantification; (3) Multi-objective optimization model; (4) Solution method. The detailed description is read in Fig. 1.

2.1. Model of the system

As shown in Fig. 1, for the model of the MCIES, the functions of the designed MCIES should be clarified first. Next, we should understand which energy sources around the building need to be applied. Then, the suitable energy conversion components are selected to convert the available energy sources into the required form of energy. Finally, energy storage devices are added to effectively smooth out the energy fluctuations, and improve the stability of the system.

Owing to the integration of multiple energy sources, the flow relationship and coupling mechanism of energy within the MCIES

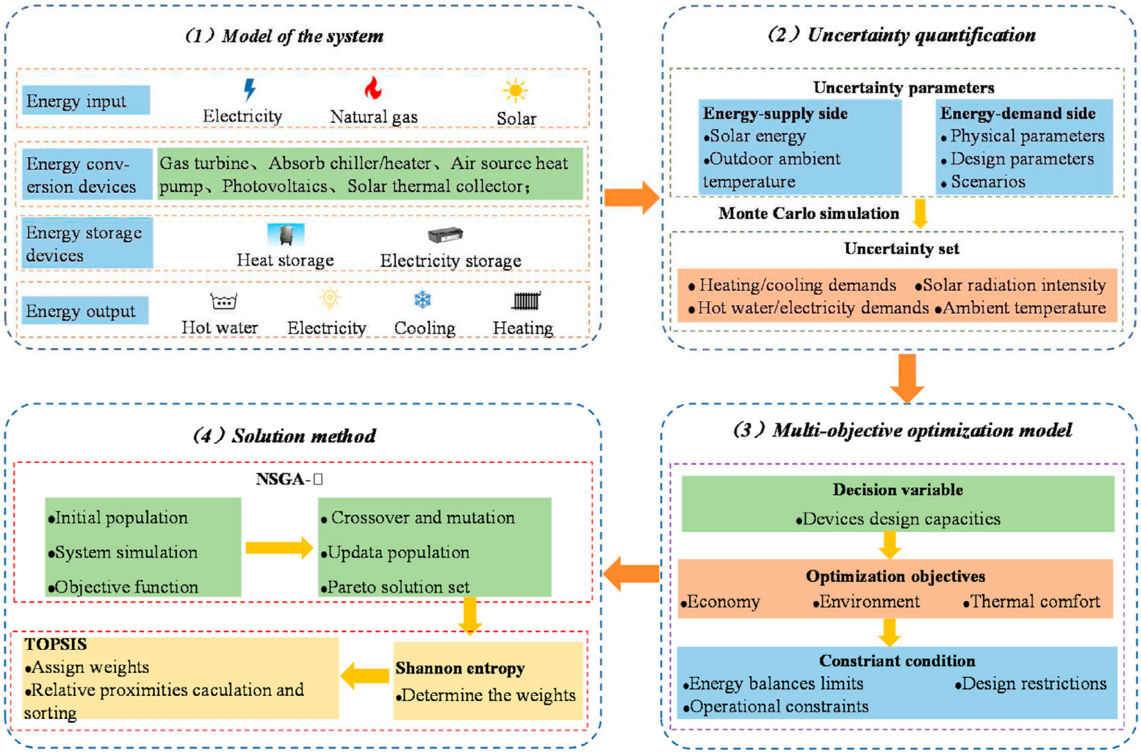

Fig. 1. The stochastic optimization framework of MCIES.

are complex and difficult to clearly describe. The energy hub (EH) model-based approach can effectively solve this problem as it clearly represents the transmission, transformation, storage and other coupling relationships between various forms of energy such as cooling, heating and electricity [34]. Based on the concept of energy hub, the EH model of a typical MCIES can be constructed as shown in Fig. 2.

Specifically, the electricity load is met first by the photovoltaic (PV), then gas turbine (GT) and finally power grid (PG). The heat of exhaust gas from GT can drive the absorption chiller/heater (AC/H) to produce cold or hot water. The hot water load and heating load are covered first by the solar thermal collector (TC), then the AC/H, followed by the air source heat pump (ASHP), and finally the electric boiler (EB). The cooling load can be satisfied first by the AC/H, followed by the ASHP. Electricity storage (ES) and thermal storage (TS) are used to store or discharge electricity and heat, respectively. The energy flow in Fig. 2 is modeled as [35]:

$$\begin{bmatrix} L_e(t) \\ L_c(t) \\ L_h(t) \\ L_{hw}(t) \end{bmatrix} = C \begin{bmatrix} P_{PG}(t) \\ V_{gas}(t) \\ Q_{aer}(t) \\ G(t) \end{bmatrix} - S \begin{bmatrix} P_{ES}(t) \\ P_{TS}(t) \end{bmatrix}, \forall t \in T \quad (1)$$

where L_e (kW) is the electricity load; L_c (kW) is the cooling load; L_h

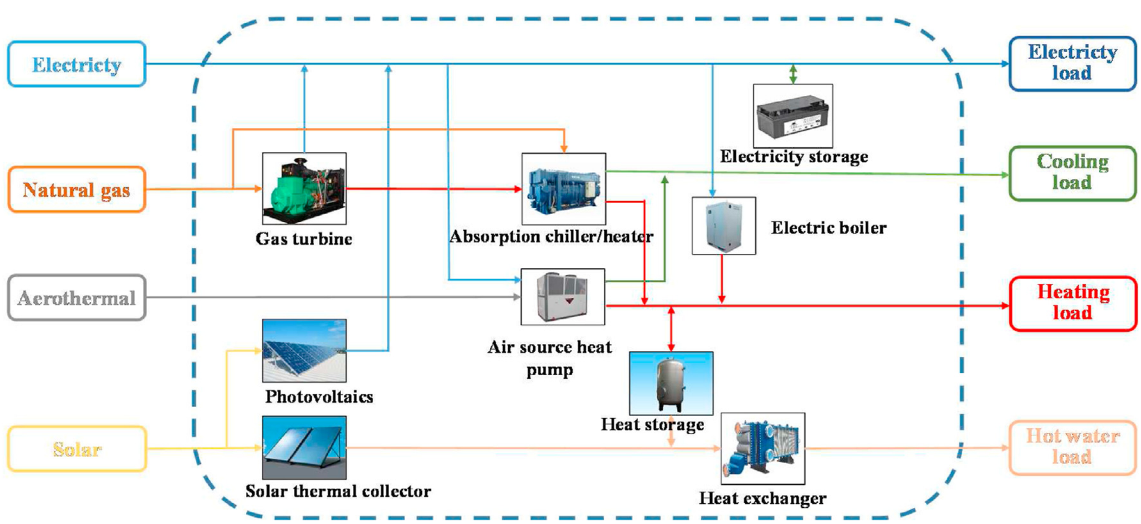

Fig. 2. Energy flowcharts of the MCIES.

(kW) is the heating load; L_{hw} (kW) is the hot water load. P_{PG} represents the output of the power grid; V_{gas} (m^3) is the gas consumption; Q_{aer} (kW) is the aerothermal energy consumption; G (W/m^2) is the solar radiation intensity. t denotes the time, and T is the set of all hours in the simulation period. P_{ES} (kW) and P_{TS} (kW) are the hourly electricity and heat energy that the battery and water tank need to charge or discharge at time period t, respectively. C and S are the converter and storage coupling matrices, respectively, which are calculated according to the mathematical models and presented in Eq. (15) and Eq. (16) respectively.

2.1.1. Gas turbine

The gas turbine (GT) using natural gas can generate electricity and heat at the same time. The equations for power generation and thermal power production are [36]:

$$P_{GT}(t) = V_{gas}(t)\eta_{GT} LHV \tag{2}$$

$$Q_{GT}(t) = \frac{P_{GT}(t)}{\eta_{GT}}(1 - \eta_{GT} - \eta_L) \tag{3}$$

where P_{GT} (kW) is the generation power of GT; Q_{GT} (kW) is the thermal power production of the GT; η_{GT} is the generation efficiency of the GT; η_L is the rate of the heat loss of the GT; LHV (kW/m^3) is the low calorific value of natural gas.

2.1.2. Absorption chiller/heater

The absorption chiller/heater (AC/H) absorbs the heat of exhaust gas from GT to produce cold or hot water, or it can burn natural gas directly for output. The output of AC/H is [18]:

$$P_{AC/H}(t) = COP_{AC/H}\left(Q_{GT}(t) + V_{gas}(t)\eta_{AC/H} LHV\right) \tag{4}$$

where $P_{AC/H}$ (kW) is the output of the AC/H; $COP_{AC/H}$ is the coefficient of performance (COP) of the AC/H and $\eta_{AC/H}$ is the thermal efficiency of natural gas combustion.

2.1.3. Air source heat pump

The output power of air source heat pump (ASHP) depends on the coefficient of performance and load profile, which can be formulated as follows:

$$P_{ASHP}^{out}(t) = P_{ASHP,e}^{in}(t) COP_{ASHP}(t) \tag{5}$$

where P_{ASHP}^{out} (kW) is the output power of ASHP; $P_{ASHP,c}^{in}$ (kW) is the power consumption of ASHP; COP_{ASHP} is the coefficient of performance of ASHP, which can be calculated as [37]:

$$COP_{ASHP}(t) = PLF(t) COP_{ASHP,d} \tag{6}$$

where PLF is the part load factor; $COP_{ASHP,d}$ is the coefficient of performance under rated working conditions, which is constant from the release data of actual product. The PLF can be calculated according to the different operating characteristics of ASHP [38]:

$$Cooling\ mode: PLF(t) = PLR(t)/(0.83PLR(t) + 0.17) \tag{7}$$

$$Heating\ mode: PLF(t) = PLR(t)/(0.73PLR(t) + 0.27) \tag{8}$$

where PLR is the part load ratio, defined as [39]:

$$PLR(t) = \frac{P_{ASHP}^{out}(t)}{P_{ASHP,d}} \tag{9}$$

where $P_{ASHP,d}$ (kW) is the rated capacity of ASHP.

2.1.4. Electric boiler

The electric boiler (EB) produces hot water by consuming electricity, which is defined as:

$$P_{EB}^{out}(t) = P_{EB}^{in}(t)\eta_{EB} \tag{10}$$

where P_{EB}^{in} (kW) is the input electric power of the EB; P_{EB}^{out} (kW) is the output thermal power of the EB; η_{EB} is the energy production coefficient of the EB.

2.1.5. Photovoltaic panels

The photovoltaic (PV) power generation depends on the external solar radiation intensity, as well as the installation area and energy conversion efficiency of the installed PV panel, formulated as follows:

$$P_{PV,e}^{out}(t) = G(t) A_{PV}\eta_{PV}(t) \tag{11}$$

where $P_{PV,e}^{out}$ (kW) is the output power of the PV panels; A_{PV} (m^2) is the installation area of the PV panels; η_{PV} is the power generation efficiency of the PV panel.

The energy conversion efficiency of the panel varies with the external solar radiation intensity and outdoor ambient temperature, which can be calculated as [40]:

$$\begin{aligned}\eta_{PV}\left(t\right) = a_1\left[\left(\frac{G(t)}{G_0}\right)^{a_2} + a_3\left(\frac{G(t)}{G_0}\right)\right] \\ \times\left[1 + a_4\left(\frac{T_{out}(t)}{T_0}\right)^{a_2} + a_5\left(\frac{AM}{AM_0}\right)\right]\end{aligned} \tag{12}$$

where T_{out} (°C) is the outdoor ambient temperature; AM (kg) is the air mass; G_0 (kW/m^2), T_0 (°C) and AM_0 (kg) are the solar radiation intensity, outdoor ambient temperature and air mass under standard test conditions, which are 1000 W/m^2, 25 °C and 1.5 kg, respectively; a_1, a_2, a_3, a_4, and a_5 are constant coefficient, which are 0.28204, 0.39668, −0.44730, −0.092864 and 0.016010, respectively [41].

2.1.6. Solar thermal collector

The output power of the solar thermal collector (TC) depends on the installation area, energy conversion efficiency and intensity of external solar radiation. The efficiency of the collector is related to the solar radiation and parameters of the collector [40].

$$\eta_{TC}(t) = F_R\tau\kappa - \frac{F_R U(T_{in} - T_{out}(t))}{G(t)} \tag{13}$$

where η_{TC} is the efficiency of TC; F_R is the heat transfer factor; τ is the transmission rate; κ is the absorption rate; U (W/($m^2 \cdot$°C)) is the heat loss coefficient of heat absorption plate; T_{out} (°C) is the outdoor ambient temperature; T_{in} (°C) is the collector water inlet temperature.

2.1.7. Energy storage device

The mathematical model of energy storage devices is shown as follows [42] :

$$E_k(t+\Delta t)=\begin{cases} E_k(t)+\int\limits_t^{t+\Delta t} P_{k,c}(t)\eta_{k,c}dt & \text{if charging mode} \\ E_k(t)-\int\limits_t^{t+\Delta t} P_{k,d}(t)/\eta_{k,d}dt & \text{if discharging mode} \end{cases} \tag{14}$$

where E_k (t) (kWh) and E_k $(t+\Delta t)$ (kWh) indicate the energy state before and after charging and discharging; $\eta_{k,c}$ and $\eta_{k,d}$ represent the charging efficiency and discharging efficiency of energy storage device k, respectively; $P_{k,c}$ (kW) and $P_{k,d}$ (kW) are the needed charging and discharging energy during period t, respectively; Δt (h) is the time interval, which is taken as 1 h in this paper.

Thus, the matrix of the converter C in Eq. (1) can be obtained from the above equations:

$$C=\begin{bmatrix} 1 & \eta_{GT}LHV & 0 & A_{PV}\eta_{PV}(t) \\ COP^C_{ASHP}(t) & COP^C_{AC/H}LHV\left(1-\eta_{GT}-\eta_L+\eta^C_{AC/H}\right) & 0 & 0 \\ \eta_{EB}+COP^H_{ASHP}(t) & COP^H_{AC/H}LHV\left(1-\eta_{GT}-\eta_L+\eta^H_{AC/H}\right) & 0 & A_{TC}\eta_{TC}(t) \\ \eta_{EB}+COP^H_{ASHP}(t) & COP^H_{AC/H}LHV\left(1-\eta_{GT}-\eta_L+\eta^H_{AC/H}\right) & 0 & A_{TC}\eta_{TC}(t) \end{bmatrix} \tag{15}$$

Noting that the heating or cooling production from the air source heat pump is a function of its electricity consumption. The relationship can be represented in column 1 of matrix C. Therefore, the coefficient related to Q_{aer} is 0 in column 3 of matrix C. Then, the matrix of S can be written as:

$$S=\begin{bmatrix} \frac{1}{\eta_{ES}} & 0 \\ 0 & 0 \\ 0 & \frac{1}{\eta_{TS}} \\ 0 & \frac{1}{\eta_{TS}} \end{bmatrix} \tag{16}$$

where η_{ES} and η_{TS} are the charge and discharge energy efficiencies of electricity and heat, respectively.

2.2. Uncertainty qualification

As shown in Fig. 1, the uncertainty sources in the sizing of MCIES are on both sides of energy production of energy-supply and load prediction of energy-demand. The uncertainties can be divided into three categories: physical parameters, design parameters and scenario parameters. For example, the weather conditions, such as solar radiation and outdoor ambient temperature, classified as the scenario parameters, will affect the energy production of PV and TC as well as the heating/cooling load prediction. Uncertainty parameters can be quantified by probability distributions [43], and a detailed description of uncertainty parameters can be found in Ref. [28]. The normal distribution is commonly used, which is expressed as follows:

$$f(x,\mu,\sigma)=\frac{1}{\sigma\sqrt{2\pi}}\exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \tag{17}$$

where x is the random variable; μ is the mean value of the random variable; σ is the standard deviation.

Fig. 3 shows the process of uncertainty handling for input parameters in system simulation. First, the uncertainties in physical parameters, design parameters and scenario parameters, are considered and quantified, which can be modeled using statistical distributions, such as normal distributions. The physical parameters are related to the physical properties of building materials, such as the density, thermal conductivity and heat capacity of walls, roofs and windows, etc. The design parameters are relative to parameters selection at the design stage, which are determined according to the design standards, such as the temperature set-point, number of air changes, etc. The scenario parameters are relative to the actual operation scenarios of the building, such as the outdoor meteorological data (i.e. ambient temperature and relative humidity, solar radiation), casual heat gain (i.e. the number of occupants, computers and light in use), etc. Subsequently, based on the standard calculation procedure of EnergyPlus software, the loads (i.e. electricity, cooling, heating and hot water load) are predicted using stochastic inputs and the Monte-Carlo simulation. The Monte-Carlo simulation is a sample-based method using random input samples to evaluate the output by repeating the simulation [44]. The outputs from the Monte-Carlo simulation are investigated by statistical analysis. The uncertainty in parameters is propagated to the loads. Then, the characteristic of the predicted loads, such as range and occurring probability, can all be obtained. Finally, loads and weather conditions with statistical distributions, constitute the uncertain input parameters of the system simulation.

2.3. Multi-objective optimization model

The problem to be solved in this paper is the proper sizing of the MCIES, i.e. how to determine the appropriate capacity of each device in MCIES. Therefore, the decision variables in the multi-objective optimization model are the rated capacity of gas turbine, absorption chiller/heater, the rated cooling and heating capacity of the air source heat pump, the rated capacity of electric boiler, installation area of photovoltaic panels and flat plate solar collectors, heat exchanger, hot water storage tank and battery. The objective function of the multi-objective optimization model is established with the system economy, environment and thermal comfort as the optimization objectives. The economic objective is to maximize the asset utilization efficiency (AUE); the environmental objective is to minimize the annual CO_2 emissions (ATE) of the system; the thermal comfort objective is to minimize the average daily unsatisfied load hours ($\Psi_{comfort}$).

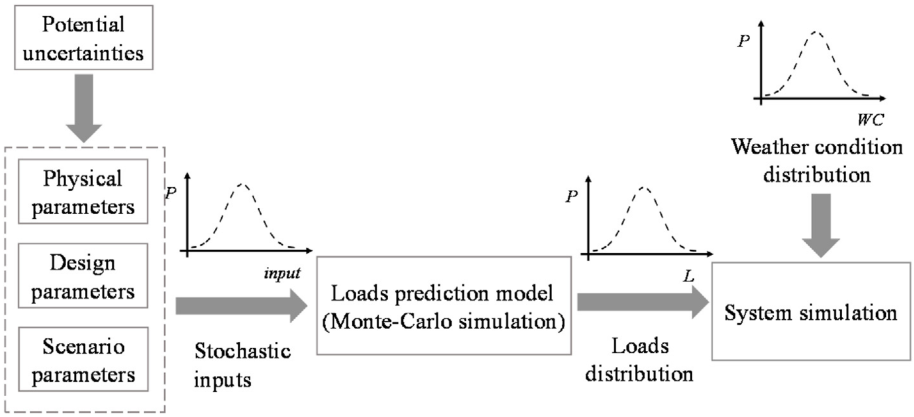

Fig. 3. Process of uncertainty handling for input parameters in system simulation.

As mentioned before, there are usually complex and contradictory relationships among these objectives. It is difficult to reflect the real situation of the system using a single objective. Therefore, a multi-objective optimization model was developed with the objectives of the system economy, environment and thermal comfort in the following form :

$$\begin{cases} \min f(X) = \min[f_1(X), f_2(X), f_3(X)] \\ subject\,to: h(X) \le 0 \\ g(X) = 0 \\ X = \left[P_{GT,d}, P_{AC/H,d}, P_{APC,d}, P_{APH,d}, P_{EB,d}, A_{PV,d}, A_{TC,d}, P_{HE,d}, E_{TS,d}, E_{ES,d}\right] \\ X \in \Omega \end{cases} \tag{18}$$

where $f(X)$ denotes the objective function, explained in detail in Subsection 2.3.1; In order to be numerically consistent and easy to calculate, the maximum asset utilization efficiency is converted to the minimum asset utilization efficiency in a negative term, i.e. $f_1(X)$ = -*AUE*, $f_2(X)$ = *ATE*, $f_3(X) = \Psi_{comfort}$. X represents the set composed of decision variables, i.e. capacities of the devices; $h(X)$ denotes the inequality constraint, detailed in Subsection 2.3.2.1; $g(X)$ denotes the equation constraint, and Ω denotes the decision space, as shown in Subsection 2.3.2.2.

2.3.1. Optimization objectives

2.3.1.1. Economic objective. The economic benefit of MCIES can be represented using asset utilization efficiency (AUE). The AUE is defined as the ratio of the saved operating cost to the investment cost of the component in a given year [45] :

$$AUE = \frac{\sum_{y=1}^{NT} \frac{1}{(1+\delta)^{y-1}} \sum_{s=1}^{NS} d_s \sum_{t=1}^{24} \left(OC_{0,y,s,t}(t) - OC^*_{y,s,t}(t)\right)}{C^T P^*} \tag{19}$$

where $OC^*_{y,s,t}$ ($) and P^* represent the operating cost and system capacity of the optimal solution; C^T ($/kW) represents the unit capacity investment cost of types of components; $OC_{0,y,s,t}$ ($) is the operating cost before the component investment is carried out; NT (year) indicates the service life of the component, which is taken as 20 years in this paper; δ represents the interest rate which is specified as 6% in this paper [46]. S and NS represent the scenario number and the total number of scenarios, respectively; d_S indicates the number of days of the scenario. The greater the AUE value, the better the economic benefits, and $f_1(X)$ = -AUE.

The cooling and heating loads are supplied through the user-side electric refrigeration/heating component (e.g., air conditioning). Their average energy efficiency coefficients are COP^h_{AC} and COP^c_{AC} respectively. Therefore, the original operating cost can be calculated as [45]:

$$OC_{0,y,s,t}(t) = ep_h(t) \times \left(L_e(t) + \frac{L_h(t) + L_{hw}(t)}{COP^h_{AC}} + \frac{L_c(t)}{COP^c_{AC}}\right) \tag{20}$$

where ep_h ($/kWh) is the hourly electricity price.

2.3.1.2. Environmental objective. Annual total CO_2 emissions are selected as the environmental objective, expressed as follows:

$$ATE = \sum_{t=1}^{T} \left(\theta_{pg} H_{pg}(t) + \theta_{ng} H_{ng}(t)\right) \tag{21}$$

where θ_{pg} ($kgCO_2/kWh$) and θ_{ng} ($kgCO_2/kWh$) represent the equivalent emission coefficient of electricity and natural gas, which are 0.968 $kgCO_2/kWh$ and 0.22 $kgCO_2/kWh$ [41], respectively;

H_{pg} (kWh) and H_{ng} (kWh) represent the grid power purchase and natural gas consumption, respectively. T (h) is taken as one year, 8760 h. It is easy to find that the smaller the ATE value, the better the environmental benefits, and $f_2(X)$ = ATE.

2.3.1.3. Thermal comfort objective. Average daily unsatisfied load hours is often applied as a thermal comfort index, which is defined as [17] :

$$\Psi_{comfort} = \left(\sum_{j=1}^{T} \tau_j\right) \Big/ 365 \begin{cases} \tau_j = 1, & if\ Q_{sup,j} \ge L_{k,j} \\ \tau_j = 0, & if\ Q_{sup,j} < L_{k,j} \end{cases} \tag{22}$$

where τ_j (h) indicates that the load in the j hour does not meet the value of the hour; $Q_{sup,j}$ (kW) is the amount of cooling or heat supplied per hour; $L_{k,j}$ (kW) is the load per hour. The smaller the $\Psi_{comfort}$ value, the better the thermal comfort, and $f_3(X) = \Psi_{comfort}$.

2.3.2. Constraints

The constraints of the MCIES optimal design problem are mainly subject to energy balance, device capacity design, and device operation constraints.

2.3.2.1. Energy balance constraints. The energy balance includes constraints on electricity, heating, cooling and hot water as equation constraints. The electricity load and power consumption of devices are met by the photovoltaic (PV), gas turbine (GT), the power grid (PG), and finally the electricity storage (ES). The air conditioning heating load and the hot water load are shared by the air source heat pump (ASHP), absorption chiller/heater (AC/H), solar thermal collector (TC), electric boiler (EB), and thermal storage (TS). The cooling load can be satisfied by the AC/H and the ASHP. These constraints can be formulated as:

$$\begin{cases} P_{PV}(t) + P_{GT}(t) + P_{PG}(t) + P_{ES}(t) = P_{ASHP,e}^{in}(t) + P_{EB}^{in}(t) + L_e(t) \\ P_{APH}(t) + P_{AH}(t) + P_{TC}(t) + P_{TS}(t) + P_{EB}(t) = L_h(t) + L_{hw}(t) \\ P_{APC}(t) + P_{AC}(t) = L_c(t) \end{cases} \tag{23}$$

where P_{APH} (kW) and P_{APC} (kW) indicate the outputs of ASHP in heating and cooling modes, respectively; P_{AH} (kW) and P_{AC} (kW) are the outputs in hot and cold water preparation mode for AC/H, respectively.

2.3.2.2. Design constraints. The design capacity range of the system component should be between the specified maximum and minimum values, which is expressed as:

$$P_{k,d}^{\min} \le P_{k,d} \le P_{k,d}^{\max} \tag{24}$$

where $P_{k,d}^{\min}$ (kW) and $P_{k,d}^{\max}$ (kW) represent the minimum and maximum capacity of device k, respectively.

For example, for the solar thermal collector (TC) and photovoltaic (PV), the environmental conditions of the building need to be met and the maximum installation area of the building cannot be exceeded, expressed as:

$$0 \le A_{TC,d} + A_{PV,d} \le A_{\max} \tag{25}$$

2.3.2.3. Operating constraints. The output power of each device during system operation can not exceed its rated power, which can be expressed as:

$$0 \le P_k^{out}(t) \le P_{k,d} \tag{26}$$

Assuming that the energy storage devices can not charge and discharge energy at the same time. Meanwhile, the charge and discharge power of energy storage devices cannot exceed its maximum limit, defined as:

$$0 \le P_k^{in}(t) \le \mu_k \alpha_{k,c}^{\max} E_{k,d}/3600 \tag{27}$$

$$0 \le P_k^{out}(t) \le (1-\mu_k) \alpha_{k,d}^{\max} E_{k,d}/3600 \tag{28}$$

where $\alpha_{k,c}^{\max}$ and $\alpha_{k,d}^{\max}$ denote the ratio of the maximum charge and discharge energy rate of the energy storage device, respectively; μ_k is a binary variable, 1 for charge mode and 0 for discharge mode.

In order to protect the energy storage device, the energy storage cannot be fully stored or empty. Thus, the stored energy needs to be maintained within a reasonable range at any time:

$$\beta_k^{\min} E_{k,d} \le E_k(t) \le \beta_k^{\max} E_{k,d} \tag{29}$$

where $\beta_k^{\min}$ and $\beta_k^{\max}$ denote the ratio of the minimum and maximum stored energy in the energy storage device, respectively.

2.4. Solution method

The solution procedure of the multi-objective optimization problem considering uncertainty is shown in Fig. 4, which is divided into three steps: (1) System simulation with uncertainty processing; (2) Pareto solution set based on the NSGA-II; (3) Decision-making using TOPSIS and Shannon entropy method.

At the uncertainty processing stage, the uncertain input parameters include solar radiation, outdoor temperature, and loads, which are randomly generated by Monte Carlo simulation. Note that the values of optimization objectives are subject to the input parameter sets. The ensemble of different optimization objectives is the so-called uncertainty set of optimization objectives. It is a challenge to select the current best solution at each iteration since each optimization objective is a statistical distribution rather than a certain value. The means are used as a representative for the next calculation.

At the Pareto solution set stage, to obtain the Pareto set of the multi-objective optimization problem, NSGA-II is used to search and calculate with better mean values of the optimization objectives as the target [11]. Each solution in the Pareto solution set consists of the values of the decision variables (X) and the optimization objectives ($f(X)$). The basic idea of NSGA-II is given in Appendix A.

In the decision-making stage, the TOPSIS combined with Shannon entropy method is used to determine the optimal solution from the Pareto solution set. The basic principle of TOPSIS is that the optimal solution has the minimum distance from the positive ideal solution and the maximum distance from the negative ideal solution [47]. Therein, the positive ideal solution is considered the best case, while the negative ideal solution is considered the worst possible case. Shannon entropy method is an objective weighting method. It relies only on the discrete nature of the data itself, rather than on the competent decision maker's weighting. The relative proximity (C_α^*) of each optimum solution in the Pareto solution set is calculated and the alternative with the maximal relative proximity is picked out as the final optimal solution ($[X_{op}, f_1\ (X_{op}), f_2\ (X_{op}), f_3\ (X_{op})]$). The steps of the TOPSIS combined with Shannon entropy method are also given in Appendix A.

3. Case study

The MCIES can be applied to many types of buildings, with the exact capacity of the equipment mainly determined by the size of the building loads. In this paper, a case study was conducted by introducing a Renewable Energy Demonstration Center (REDC) located in Changsha city, China, as the prototype was used to illustrate the design procedure of the proposed multi-objective optimization method for MCIES considering load prediction and energy production uncertainties. Fig. 5 shows the aerial view of the REDC. The main body of the REDC is an indoor swimming pool with a total surface area of 450 m^2 and a volume of 540 m^3. The maximum area of the swimming pool roof available for the installation of PV and TC panels is 372 m^2. The swimming pool is open from 9 a.m. to 10 p.m. every day.

3.1. Uncertainty description and parameter setting

3.1.1. Uncertainty description

The uncertainty parameters considered in this study are summarized in Table 1, assumed to comply with the normal distribution. In the table, N (a, b) denotes the mathematical expectation of a, the variance of b. Table 2 displays 12 typical scenarios and the probability of each scenario. The solar radiation temperature data are obtained from the typical meteorological year (TMY) data [48] downloaded from the Meteonorm software. And the probability of weather scenarios is calculated based on the statistics from the

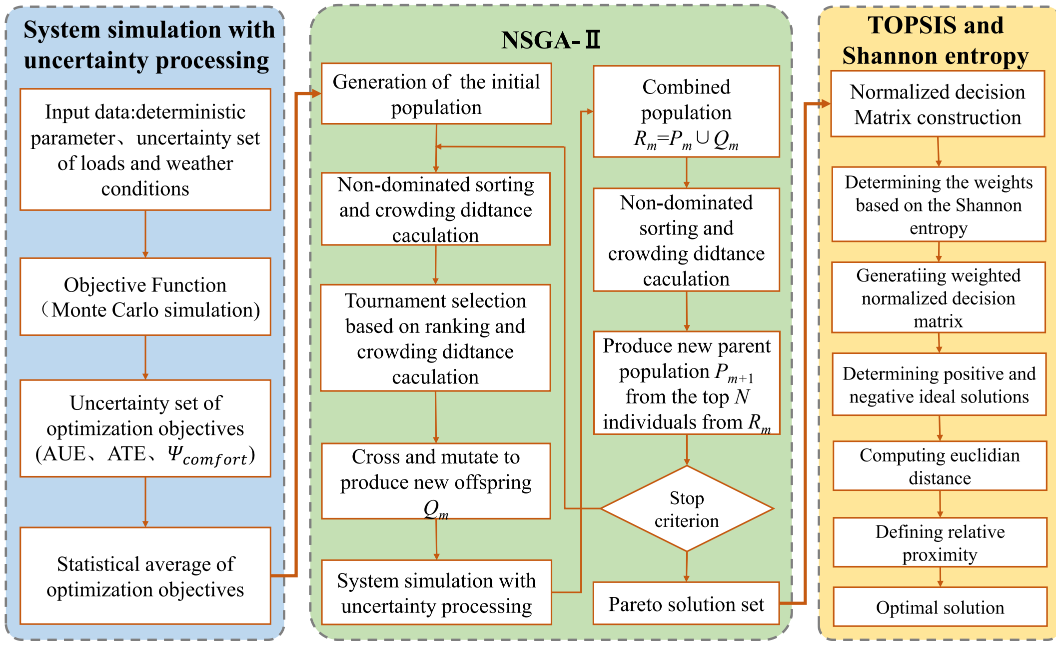

Fig. 4. Multi-objective optimization method considering uncertainty.

Fig. 5. Aerial view of the REDC in Changsha.

website [49]. The 12 scenarios are assumed to have different solar radiation values, while the outdoor temperatures and loads are the same for the same seasonal scenarios. The distributions of the outdoor temperature are given in Fig. B1. The hourly solar radiation is displayed in Fig. B2. As mentioned in Subsection 2.2, the distributions of hourly load prediction are obtained using the Monte Carlo simulation, including space cooling/heating load, electricity load, and hot water load for the swimming pool. The load distributions are shown in Figs. B3-B5 of Appendix B.

Table 2
Probability of different scenarios [49].

	Cloudy day	Rainy day	Clear day
Spring	0.085	0.101	0.025
Summer	0.110	0.096	0.074
Autumn	0.088	0.041	0.036
Winter	0.175	0.101	0.068

3.1.2. Parameter setting

The parameters setting in simulation cases are introduced in this subsection. Table 3 shows the initial investment and service lifetime of the key components of the MCIES, which are used to calculate the economic objective AUE of the MCIES. The technical parameters of some equipment are summarized in Table 4. Table 5 shows the performance parameters of the energy storage devices. The prices information of natural gas and electricity used in this case study are listed in Table 6. The relevant parameters setting of the NSGA-II is presented in Table 7. To better allocate the optimization space, except for the PV and TC, the minimum value of the

Table 1
The description of the uncertainty parameters [17,28].

Parameter	Base value	Distribution	Truncation
Occupant number	240	N (240, 12^2)	[0, 240]
Surface wind velocity (m/s)	0.2	N (0.2, 0.02^2)	[0, 0.3]
Equipment gains (W/m^2)	20	N (20, 1^2)	[15, 25]
Supplementary water temperature (°C)	T_f	N (mean, 2^2)	–
Ambient temperature (°C)	T_a	N (mean, 2^2)	–
Solar radiation (W/m^2)	I_a	N (mean, σ_1^2), $\sigma_1 = 0.12$, mean (winter, spring, 9:00–15:00)	–
		N (mean, σ_2^2), $\sigma_2 = 0.25$, mean (winter, spring, other time)	
		N (mean, σ_3^2), $\sigma_3 = 0.03$, mean (summer, Autumn, 9:00–15:00)	
		N (mean, σ_4^2), $\sigma_4 = 0.08$, mean (summer, Autumn, other time)	

Table 3
The economic parameters of the key MCIES components [41,50].

Component	Investment cost	Lifetime
GT	409.8 \$/kW	20
AC/H	157.5 \$/kW	20
ASHP	183.5 \$/kW	20
EB	132.4 \$/kW	20
PV	283.5 \$/m^2	20
TC	255.6 \$/m^2	20
TS	14.0 \$/kWh	20
ES	281.0 \$/kWh	13.5
HE	31.3 \$/kW	20

Table 4
Technical parameters of MCIES [36,51].

Equipment	Parameter	Value
GT	η_{GT}	0.3
	η_L	0.08
	LHV, kW/m^3	9.7
AC/H	$\eta_{AC/H}$	0.98
	$COP^C_{AC/H}$	0.9
	$COP^H_{AC/H}$	1.4
EB	η_{EB}	0.9
TC	F_R	0.748
	τ	0.95
	k	0.94
	U, W/(m$^2\cdot$°C)	5.4
	T_{in}, °C	40
PV	AM, kg	2.5

capacity of other devices in NSGA-II is 0 and the maximum value is 1.5 times the load they need to satisfy.

3.2. Results and discussions

In this subsection, the simulation results for the comparison of the proposed uncertainty-based multi-objective stochastic optimal design method (UMSODM) and the conventional deterministic optimal design method (DODM) cases were first given. Then, we analyzed the impacts of uncertainty degrees and scenario settings on the performance of the proposed uncertainty-based multi-objective stochastic optimal design method, respectively. Note that all simulations were taken using MATLAB (2018b, The MathWorks, Inc., Massachusetts, USA) software installed on a computer with a Quadcore Intel(R) Core(TM) i7-10750H CPU @ 2.60 GHz 2.59 GHz and 16 GB of RAM. To allow readers to better follow and understand the results, some further explanations are given as follows:

(1) The calculation procedure for comparison of both methods is shown in Fig. 6. The two Pareto solution sets were first merged into one solution set. Then, the TOPSIS combined with Shannon entropy method, introduced in Subsection 2.4, was used to calculate the relative proximity C* for each solution. Finally, the relative proximities for the Pareto solution sets of the deterministic optimal design method (C^*_{Dop}) and the uncertainty-based multi-objective stochastic optimal design method (C^*_{Uop}) were ranked. The larger the relative proximity of the alternative, the better is the performance of the design method.

(2) To investigate the effects of uncertainty degrees on the MCIES capacity optimization, we designed five types of uncertainty degrees, which were quantified using different standard deviations of the uncertainty parameters. The degree of uncertainty was denoted by the variable x, which was set to 0, 0.5, 1, 1.5, 2. The higher value of x denoted that the uncertainty degree of the designed case was larger. For example, in case $x = 1.5$, the standard deviations of the relative uncertainty parameters were 1.5 times larger than that in the case with $x = 0$. Therein, the uncertain distributions of the parameters in case $x = 1$ were shown in Table 1.

Table 6
The prices of natural gas and electricity [41].

Item	Unit	Peak hours	High hours	Flat hours	Valley hours
		19–21	8–10, 15–18	7, 11–14, 22	1–6, 23–24
Electricity	\$/kWh	0.1279	0.1134	0.0916	0.0625
Natural gas	\$/m^3	0.4840	04840	0.4840	0.4840

Table 7
The parameters setting for NSGA-II.

Parameters	Value
Decision variable	10
Objective function	3
Population size	200
Generation time	500
Crossover probability	0.9
Mutual probability	0.1
Crossover distribution	20
Mutual distribution	20

3.2.1. Comparison of the uncertainty-based multi-objective stochastic optimal design method and deterministic optimal design method

The optimal capacities from the uncertainty-based multi-objective stochastic optimal design method and the deterministic optimal design method are shown in Table 8. The design capacity of GT from the uncertainty-based multi-objective stochastic optimal design method is 10 kW, smaller than that from the deterministic optimal design method. While the design capacities of APH and APC from uncertainty-based multi-objective stochastic optimal design method are 190 kW and 88 kW, respectively, much larger than those for the deterministic optimal design method. The MCIES

Table 8
The optimal capacities based on the uncertainty-based multi-objective stochastic optimal design method (UMSODM) and the deterministic optimal design method (DODM).

Methods	$P_{GT,d}$	$A_{PV,d}$	$A_{TC,d}$	$P_{APH,d}$	$P_{APC,d}$	$P_{HE,d}$	$E_{TS,d}$	$E_{ES,d}$	$P_{EB,d}$	$P_{AC/H,d}$
	kW	m^2	m^2	kW	kW	kW	kWh	kWh	kW	kW
UMSODM	10	75	297	190	88	156	185	16	65	29/18
DODM	20	22	350	162	80	155	165	11	64	58/36

Table 5
The technical parameters of energy storage devices [52].

Component	Maximum charging ratio	Maximum discharging ratio	Minimal stored energy ratio	Maximal stored energy ratio	Charge efficiency	Discharge efficiency
ES	0.30	0.35	0.20	0.90	0.96	0.96
TS	0.40	0.40	0.20	0.90	0.98	0.98

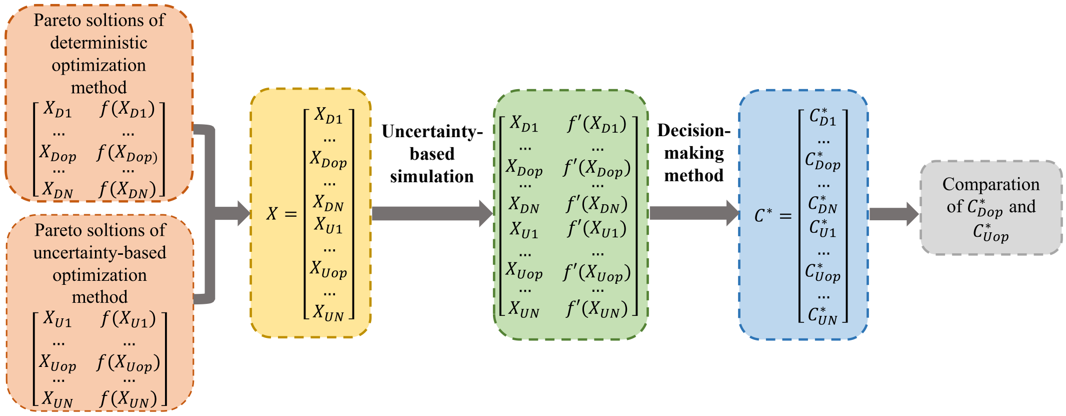

Fig. 6. The calculation procedure for comparison of both methods.

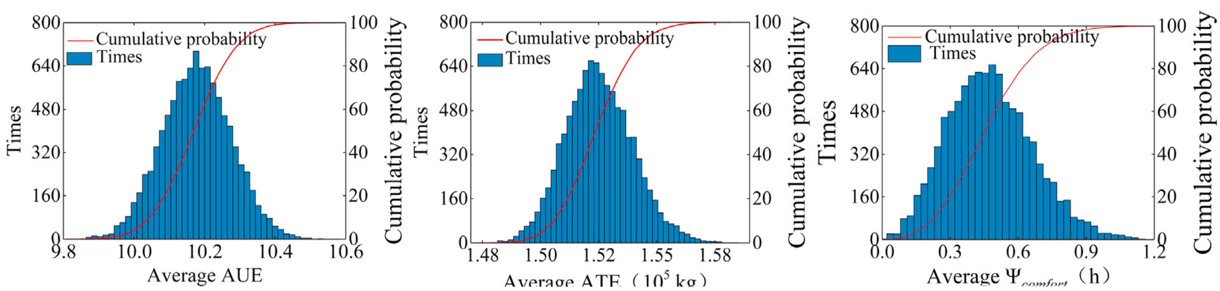

Fig. 7. Histograms of AUE, ATE and $\Psi_{comfort}$ under the optimal capacity of the uncertainty-based multi-objective stochastic optimal design method.

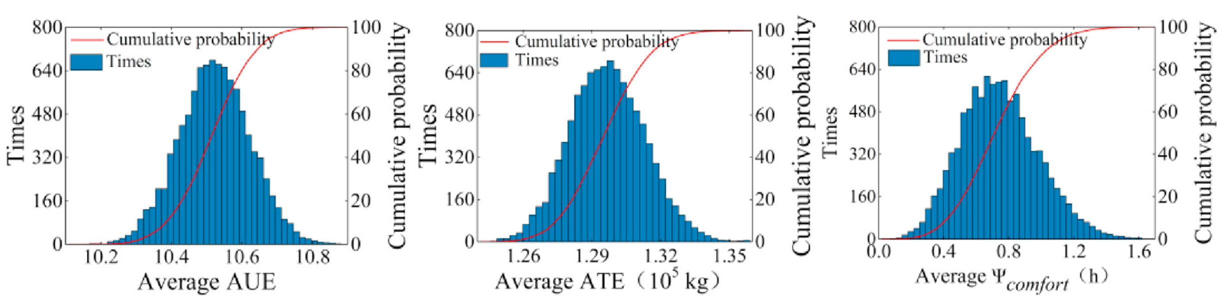

Fig. 8. Histograms of AUE, ATE and $\Psi_{comfort}$ under the optimal capacity of the deterministic optimal design method.

with larger capacities of APH and APC means that the system has better robustness to deal with uncertain conditions. This is the reason that the uncertainty-based multi-objective stochastic optimal design method can obtain better design performance.

Figs. 7 and 8 show the frequency and cumulative probability distribution of the AUE, ATE and $\Psi_{comfort}$ obtained from the optimal capacity solutions of the uncertainty-based multi-objective stochastic optimal design method (UMSODM) and the deterministic optimal design method (DODM) under the same uncertainty simulation setting. The horizontal coordinate indicates the range of variation of the optimization objective. The vertical coordinate on the left indicates the number of times the optimization objective falls in a certain range of values 10,000 times, and the one on the right indicates the cumulative probability of the optimization objective. The mean values of AUE, ATE and $\Psi_{comfort}$ are given in Table 9. Compared with the uncertainty-based multi-objective stochastic optimal design method (UMSODM), the mean values of AUE and $\Psi_{comfort}$ with the deterministic optimal design method (DODM) increase by 3.34% and 85%, respectively, and the mean value of ATE decreases by 15.03%. It shows that the deterministic optimal design method results in better economic and environmental benefits, but comparatively poorer thermal comfort benefit.

Table 9
The mean values of the optimization objectives.

Method	AUE	ATE/10^5 kg	$\Psi_{comfort}$/h
UMSODM	10.18	1.53	0.40
DODM	10.52	1.30	0.74

Fig. 9 shows the relative proximities (C^*_α) of 400 Pareto solutions of the two methods. The first two hundred C^*_α are calculated using the deterministic optimal design method's Pareto solutions, and the last two hundred C^*_α are calculated using the uncertainty-based multi-objective stochastic optimal design method's Pareto solutions. As seen in Fig. 9, the maximal relative proximity from the uncertainty-based multi-objective stochastic optimal design

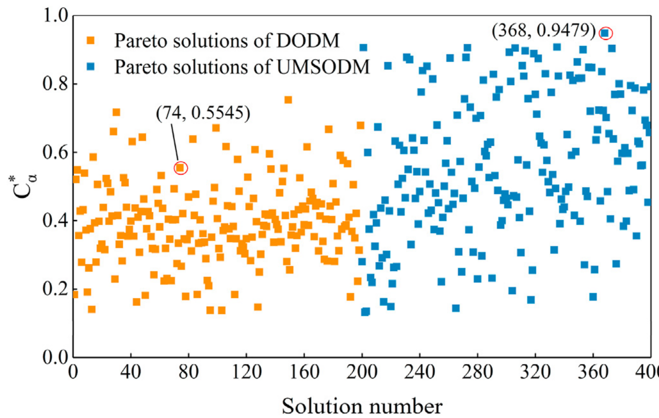

Fig. 9. The relative proximities of Pareto solution sets of the two methods.

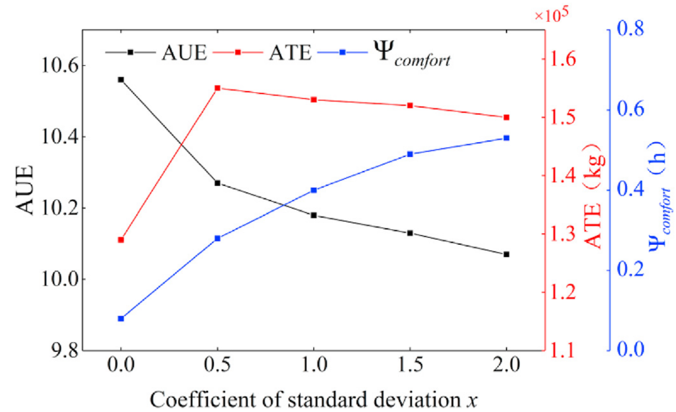

Fig. 10. The influence of uncertainty on the performance indexes.

method (C^*_{Uop}) is 0.9479, while that from the deterministic optimal design method (C^*_{Dop}) is 0.5545. It indicates that the system performance from the uncertainty-based multi-objective stochastic optimal design method is much better than that from the deterministic optimal design method, in terms of comprehensively considering the three optimization objectives.

3.2.2. Impact of uncertainty degree on the MCIES capacity optimization

Table 10 shows the influence of uncertainty degree on the capacity sizing of the selected devices and the AUE, ATE, $\Psi_{comfort}$. The case with $x = 0$ means the parameter information used is with no uncertainty, which has better optimization objectives. When x varies from 0.5 to 2, the capacities of GT, APH, HE and EB; while changes in remaining device capacities have no fixed trend. Fig. 10 represents the influence of uncertainty on the optimization objectives. As the uncertainty increases, the economic and thermal comfort benefits deteriorate, while there is a small improvement in environmental benefit, and the change ranges of the three optimization objectives decrease. In addition, $\Psi_{comfort}$ is more sensitive to uncertainty change than AUE and ATE. For example, compared with $x = 0$, when $x = 0.5$, the AUE decreases by 2.75%, ATE increases by 20.16%, while $\Psi_{comfort}$ increases by 250%. The TOPSIS and Shannon entropy methods are used to analyze five solutions, where the C^*_α of each solution is listed in Table 10. The magnitude of C^*_α can reflect the integrated system performance, and the larger the C^*_α, the better the integrated system performance. As x increases, the C^*_α decreases, reflecting that the integrated performance of MCIES deteriorates.

3.2.3. Impact of scenario setting on the MCIES capacity optimization

To investigate the impact of scenario setting on the MCIES capacity optimization, three schemes were designed. Scheme 1 is the baseline scheme with hourly simulation throughout the 365 original scenario days, using the typical meteorological year data. Scheme 2 and 3 select 12 and 4 typical scenario days to simulate the full-year to reduce the simulation time, respectively. Three schemes are described in detail as follows:

Scheme 1: Full-year simulations are performed in this case using the location's typical meteorological year (TMY) [48] data as the reference conditions.

Scheme 2: The case is calculated in a typical year divided into 12 typical scenario days. The 12 typical scenario days are selected according to the four seasons (e.g., spring, summer, autumn and winter) and three kinds of weather (e.g., clear, cloudy and rainy). Note that three typical scenario days are selected during each season with the same meteorological data, except for the solar radiation. The ambient temperature and solar radiation intensity are shown in Fig. B1 and Fig. B2 of Appendix B. The duration of each typical scenario day is shown in Table 2.

Scheme 3: The case is calculated in a typical year divided into four seasonal typical days including spring, summer, autumn and winter season. The ambient temperature data of the four seasonal

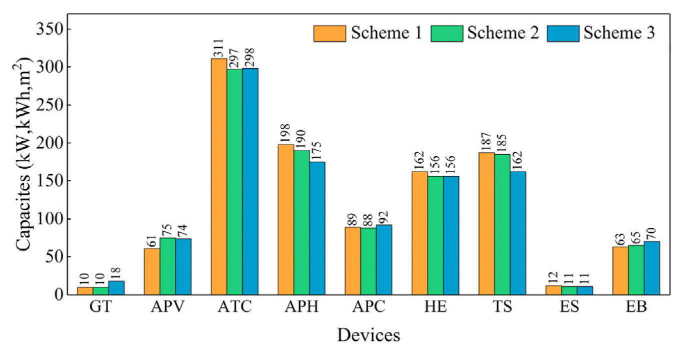

Fig. 11. The optimal capacities of Scheme 1, 2 and 3.

Table 10
The optimal solution with different x.

x	$P_{GT,d}$	$A_{PV,d}$	$A_{TC,d}$	$P_{APH,d}$	$P_{APC,d}$	$P_{HE,d}$	$E_{TS,d}$	$E_{ES,d}$	$P_{EB,d}$	$P_{AC/H,d}$	AUE	ATE	$\Psi_{comfort}$	C^*_α
	kW	m^2	m^2	kW	kW	kW	kWh	kWh	kW	kW	/	10^5 kg	h	
0	20	22	350	162	80	155	165	11	64	58/36	10.56	1.29	0.08	1
0.5	10	44	328	184	86	152	170	13	65	29/18	10.27	1.55	0.28	0.5556
1.0	10	75	297	190	88	156	185	16	65	29/18	10.18	1.53	0.40	0.2889
1.5	11	54	318	198	87	166	190	12	70	32/20	10.13	1.52	0.49	0.0889
2.0	13	35	337	200	89	176	198	11	82	38/24	10.07	1.50	0.53	0.0005

Table 11
The optimization objectives and C_α^* of Scheme 1, 2 and 3.

Schemes	AUE	ATE/10^5 kg	$\Psi_{comfort}$/h	C_α^*
1	10.24	1.54	0.48	0.9275
2	10.19	1.51	0.49	0.8962
3	10.07	1.29	0.68	0.7086

typical days are shown in Fig. B1. While the solar radiation intensity data is selected using the sunny day are shown in Fig. B2. The duration of each typical scenario day is shown in Table 2.

The calculation time of the MCIES capacity optimization for Scheme 1–3 were 233.34 h, 5.87 h and 2.23 h, respectively. The computational burden for Scheme 1 is almost 10 days, which is not affordable at the design phase. Thus, using typical scenarios instead of full-year hourly simulation calculation is an effective and applicable way. Fig. 11 shows the optimal capacities of Scheme 1, 2 and 3. The optimization objectives obtained for different device capacity configurations under the simulation conditions of Scheme 1 are shown in Table 11. The devices' capacities and optimization objectives of Scheme 2 are much closer to Scheme 1 than Scheme 3. After comparing with the Pareto solution set of Scheme 1, as seen in Table 11, the C_α^* of the three schemes are 0.9275, 0.8962 and 0.7086, respectively. The relative error of C_α^* for Scheme 1 and Scheme 2 is 3.37%, which is much smaller than the relative error of 23.60% for Scheme 1 and Scheme 3. The capacity optimization method based on only four seasonal typical days will significantly deteriorate the system performance of the MCIES. While the scheme of 12 typical scenario days can obtain a nearly optimal solution with saving nearly 40 times of calculation time. It is a suitable scenario setting strategy for the MCIES capacity optimization using the uncertainty-based stochastic optimal design method.

4. Conclusions

Considering the uncertainties in load prediction and renewable energy production, this paper proposed an uncertainty-based multi-objective stochastic optimal design method (UMSODM) for optimal sizing of MCIES. Uncertainties in load prediction and weather conditions could be quantified using statistical distributions. Monte Carlo simulation was used to obtain the uncertainty distributions of the load prediction and renewable energy production. NSGA-II was used to solve the stochastic multi-objective optimization problem. This study combined TOPSIS with Shannon entropy, as the decision strategy, to pick up the final optimal solution from the Pareto solution set. A Renewable Energy Demonstration Center (REDC) located in Changsha city, China, was used as the prototype. Simulations were carried out and the performance of this novel design method was evaluated, in comparison with the conventional deterministic optimal design method (DODM). In addition, this study also explored the impact of uncertainty degree and scenario setting. The main conclusions of this paper are as follows :

(1) Considering the uncertainties in the actual operating environment of the system, the optimal capacities derived from the uncertainty-based optimal design approach, can result in better integrated economic, environmental and thermal comfort benefits compared with the deterministic optimal design method.
(2) The uncertainties of renewable energy sources and loads have non-negligible impacts on the capacity design of the MCIES. The increase in uncertainty will lead to an increase in the capacity of some devices for system robustness. The addition in uncertainty can improve or worsen individual optimization objectives, but it will lead to deterioration of the integrated system performance.
(3) Adequate and appropriate scenario delineation plays a critical role in the uncertainty-based capacity optimization of MCIES. Considering the model solving time and the closeness of the solution, the 12-scenario setting is more suitable to replace the original 365-scenario setting than the 4-scenario setting.

The uncertainty-based multi-objective stochastic optimal design method proposed in this paper was applied based on the pre-defined established device types and determinate operation strategies. In future research, the performance of this proposed method will be further investigated with different system configurations of MCIES and the impacts of the operation strategies should be studied. Meanwhile, this paper assumed that the uncertainty parameters follow the normal distribution and other types of statistical distributions are still to be studied.

Credit author statement

Zhiqiang Liu provided the concept and revised the final version. Yanping Cui conducted the literature review, modelling and wrote the first draft of the manuscript. Jiaqiang Wang provided the conceptualization, methodology and analyzed the measured data. Chang Yue revised the original draft and edited the draft of manuscript. Yawovi Souley Agbodjan provided some models and edited the draft of manuscript. Yu Yang revised and edited the draft of manuscript. All authors replied to reviewers' comments and revised the final version.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Basic ideas and programming of the solution method

A.1 Basic ideas of the NSGA-II and TOPSIS combined with Shannon entropy.

The basic idea of NSGA-II is shown as follows [41]:

- Initialization: the parameters, including population size N, evolution generation G, crossover probability and mutual probability, are set up. The initial population, P_m, containing N chromosomes is randomly created under constraints. Then the objective functions of each individual are calculated.
- Non-dominated sorting and crowding distance calculation: the fast non-dominated sorting is implemented for the population

P_m, and the rank of each individual is assigned. Then the crowding distance of each individual is computed.
- Tournament selection: the approach of tournament selection is employed. Two individuals are randomly selected from population P_m at a time, of which the individual with top ranking and high crowding degree is picked out.
- Genetic operators: simulated binary crossover and polynomial mutation operators are used for the genetic operation to produce the new offspring, Q_m.
- Recombination: population R_m is created via combining the father population P_m with offspring population Q_m.
- Non-dominated sorting and crowding distance calculation is performed for the population R_m.
- Produce new generation population: select the top N individuals from R_m as the new generation population P_{m+1} based on their rank and crowding distance.
- Judge whether the terminal condition is met. If terminal condition is satisfied, output Pareto solution set. Otherwise, turn to Tournament selection and repeat until terminal condition is met.

The steps of the TOPSIS combined with Shannon entropy method are shown below [53] :

(1) Determining the normalized decision matrix

$$\gamma_{\alpha\beta} = \frac{f_{\alpha\beta}}{\sqrt{\sum_{\alpha=1}^{N} f_{\alpha\beta}^2}}, \alpha = 1, 2, \ldots, N; \beta = 1, 2, 3 \tag{A1}$$

where $f_{\alpha\beta}$ is the member of the decision matrix denoting the β_{th} objective value of the α_{th} optimum solution. N is the total number of optimum solutions in the Pareto solution set, which equals to population size. $\gamma_{\alpha\beta}$ represents the element of the normalized decision matrix.

(2) Determining the weights based on the Shannon entropy approach

$$O_\beta = -\frac{1}{\ln N}\sum_{\alpha=1}^{N}\left(\frac{f_{\alpha\beta}}{\sum_{\alpha=1}^{N} f_{\alpha\beta}^2}\ln\frac{f_{\alpha\beta}}{\sum_{\alpha=1}^{N} f_{\alpha\beta}^2}\right) \tag{A2}$$

$$J_\beta = \frac{1 - O_\beta}{3 - \sum_{\beta=1}^{3} O_\beta} \tag{A3}$$

(3) Generating weighted normalized decision matrix

$$\nu_{\alpha\beta} = \gamma_{\alpha\beta} J_\beta \tag{A4}$$

(4) Determining positive ideal ($\nu + \beta$) and negative ideal ($\nu - \beta$) solutions

$$\nu_\beta^+ = \left\{\left(\max_\alpha \nu_{\alpha\beta}\middle|\beta \in J_1\right), \left(\min_\alpha \nu_{\alpha\beta}\middle|\beta \in J_2\right)\right\} \tag{A5}$$

$$\nu_\beta^- = \left\{\left(\min_\alpha \nu_{\alpha\beta}\middle|\beta \in J_1\right), \left(\max_\alpha \nu_{\alpha\beta}\middle|\beta \in J_2\right)\right\} \tag{A6}$$

where J_1 and J_2 denote benefit type criteria and cost type criteria, respectively.

(5) Computing the Euclidian distance

$$Z_\alpha^+ = \sqrt{\sum_{\beta=1}^{3}\left(\nu_{\alpha\beta} - \nu_\beta^+\right)^2} \tag{A7}$$

$$Z_\alpha^- = \sqrt{\sum_{\beta=1}^{3}\left(\nu_{\alpha\beta} - \nu_\beta^-\right)^2} \tag{A8}$$

(6) Defining the relative proximity

The relative proximity C_α^* closer to 1 means that the alternative solution is closer to the positive ideal solution, i.e., the alternative is better.

$$C_\alpha^* = \frac{Z_\alpha^-}{Z_\alpha^+ + Z_\alpha^-} \tag{A9}$$

(7) Optimal solution

Ranking the alternatives: each alternative in the Pareto solution set is sorted according to the value of the C_α^*. The alternative with the maximal relative proximity is picked out as the final optimal solution.

$$X_{op} = \left\{X_i \middle| \max_{1\le i\le N}(C_\alpha^*)\right\} \tag{A10}$$

A.2 The programming of the solution method.

Set Population size $N = 200$, Evolution generation $G = 500$, Decision variable $V = 10$, Objective function $f = 3$.
%% System simulation with uncertainty processing **for i = 1:N** Randomly initialize $X_i = [P_{GT,d}, P_{AC/H,d}, P_{APC,d}, P_{APH,d}, P_{EB,d}, A_{PV,d}, A_{TC,d}, P_{HE,d}, E_{TS,d}, E_{ES,d}]$ Set the uncertainty parameters given in Tables 1–2, determined parameters given in Tables 3–6, and cooling/heating load, electricity load, and hot water load given in Figs. B1-B3. Set simulation times $M = 10{,}000$. **for i = 1:M** Solve AUE, ATE, $\Psi_{comfort}$ using Eqs. (1-16) and Eqs. (19-29) **end** $f_1(X_i) = -(\sum_{i=1}^{M} AUE)/M$ $f_2(X_i) = (\sum_{i=1}^{M} ATE)/M$, $f_3(X_i) = (\sum_{i=1}^{M} \psi_{comfort})/M$ **end** **%% Generate the initial population**

(continued on next page)

(continued)

Set Population size $N = 200$, Evolution generation $G = 500$, Decision variable $V = 10$, Objective function $f = 3$.

$P_m = \begin{bmatrix} X_1 & f_1(X_1) & f_2(X_1) & f_3(X_1) \\ \dots & \dots & \dots & \dots \\ X_N & f_1(X_N) & f_2(X_N) & f_3(X_N) \end{bmatrix}$

%% Non-dominated sorting and crowding distance calculation

Return two columns for each alternative which are the rank and the crowding distance corresponding to their position in the front they belong.

Get $P_{NS} = \begin{bmatrix} X'_1 & f_1(X'_1) & f_2(X'_1) & f_3(X'_1) \\ \dots & \dots & \dots & \dots \\ X'_N & f_1(X'_N) & f_2(X'_N) & f_3(X'_N) \end{bmatrix}$

for $i = 1{:}G$

%% Tournament selection

Select the top $N/2$ alternatives from P_{NS},

Get $P_{TS} = \begin{bmatrix} X''_1 & f_1(X''_1) & f_2(X''_1) & f_3(X''_1) \\ \dots & \dots & \dots & \dots \\ X''_{N/2} & f_1(X''_{N/2}) & f_2(X''_{N/2}) & f_3(X''_{N/2}) \end{bmatrix}$

%% Genetic operators

Set Crossover distribution $cd = 20$, Mutual distribution $md = 20$, Crossover probability $Cp = 0.9$, Mutual probability $Mp = 0.1$.

for $i = 1{:}N$

Generate $X_{i,C/M} = [P_{GT,d}, P_{AC/H,d}, P_{APC,d}, P_{APH,d}, P_{EB,d}, A_{PV,d}, A_{TC,d}, P_{HE,d}, E_{TS,d}, E_{ES,d}]$ using crossover and mutation operation

Read the uncertainty parameters and determined parameters.

for $i = 1{:}M$

Solve AUE, ATE, $\Psi_{comfort}$ using Eqs. (1-16) and Eqs. (19-29)

end

$f_1(X_{i,C/M}) = -(\sum_{i=1}^{M} AUE)/M$ $f_2(X_{i,C/M}) = (\sum_{i=1}^{M} ATE)/M$, $f_3(X_{i,C/M}) = (\sum_{i=1}^{M} \psi_{comfort})/M$

end

Produce the new offspring $Q_m = \begin{bmatrix} X_{1,C/M} & f_1(X_{1,C/M}) & f_2(X_{1,C/M}) & f_3(X_{1,C/M}) \\ \dots & \dots & \dots & \dots \\ X_{Q,C/M} & f_1(X_{Q,C/M}) & f_2(X_{Q,C/M}) & f_3(X_{Q,C/M}) \end{bmatrix}$

%% Recombination

$R_m = P_m \cup Q_m$

%% Non-dominated sorting and crowding distance calculation

Get R'_m

%% Produce new generation population

Select the top N alternatives from R'_m.

Get $P_i = \begin{bmatrix} X_1^i & f_1(X_1^i) & f_2(X_1^i) & f_3(X_1^i) \\ \dots & \dots & \dots & \dots \\ X_N^i & f_1(X_N^i) & f_2(X_N^i) & f_3(X_N^i) \end{bmatrix}$

end

Get $P_G = \begin{bmatrix} X_1^G & f_1(X_1^G) & f_2(X_1^G) & f_3(X_1^G) \\ \dots & \dots & \dots & \dots \\ X_N^G & f_1(X_N^G) & f_2(X_N^G) & f_3(X_N^G) \end{bmatrix}$

% % TOPSIS and Shannon entropy method

$F = abs \begin{bmatrix} f_1(X_1^G) & f_2(X_1^G) & f_3(X_1^G) \\ \dots & \dots & \dots \\ f_1(X_N^G) & f_2(X_N^G) & f_3(X_N^G) \end{bmatrix}$

Solve C_α^*, X_{op} and $f_1(X_{op})$, $f_2(X_{op})$, $f_3(X_{op})$ using Eqs. (A1-A10).

Appendix B. Distributions of the loads and weather parameters

Fig. B1 displays the distributions of the outdoor temperature, which is divided into four scenarios with a standard deviation of 2 °C. Fig. B2 shows the distributions of the hourly solar radiation on 12 typical days selected from the location's typical meteorological year (TMY) data. Note that the solar radiation for the hours that do not appear in Fig. B2 is 0.

Figs. B3-B5 show the load distributions. The middle lines are the means of the prediction loads, and the shades of color in the graph indicate the sparsity of the data, characterizing the likelihood of occurrence. The uncertainty of renewable energy production is subjected to the uncertainties of solar radiation and outdoor ambient temperature, which are also described using the normal distributions.

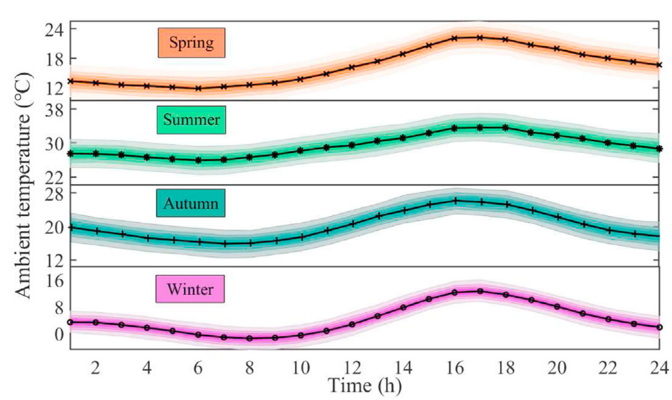

Fig. B1. Hourly ambient temperature distributions in different scenarios. .

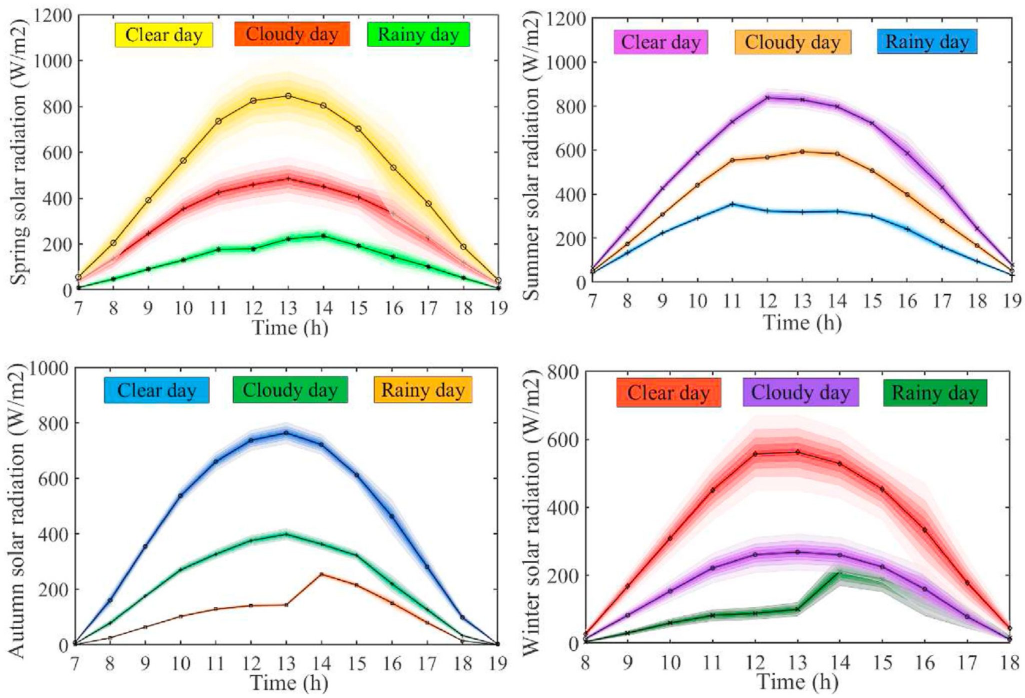

Fig. B2. Hourly solar radiation intensity distributions in different scenarios. .

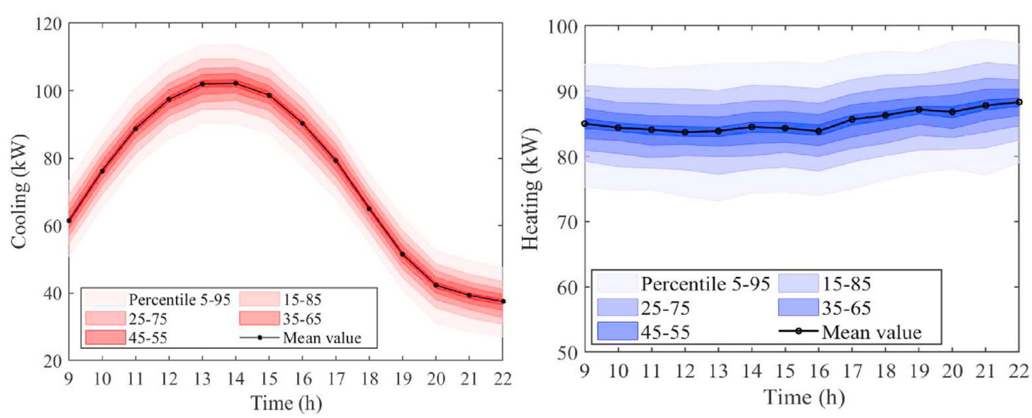

Fig. B3. Hourly cooling and heating loads prediction distributions of the REDC. .

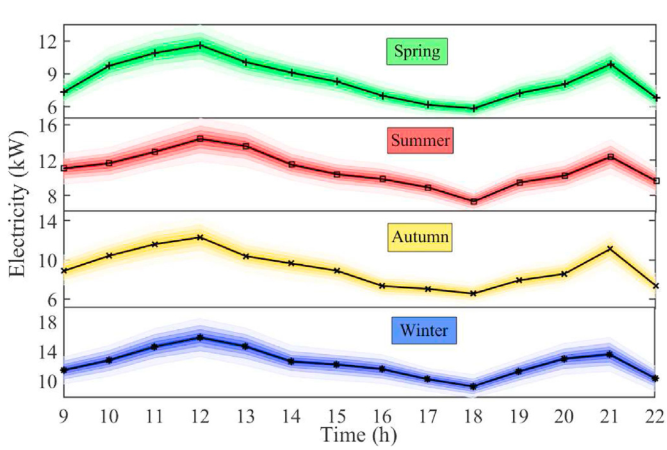

Fig. B4. Hourly electricity load prediction distributions of the REDC. .

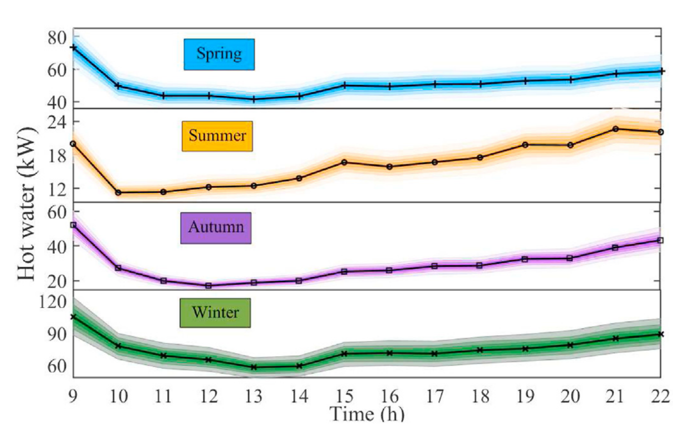

Fig. B5. Hourly hot water load prediction distributions of the REDC.

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